

HDOC DAY-37 ASSIGNMENT

Menu-Driven String Operations Using Separate Functions

1. Problem Understanding

The task is to write a menu-driven C program that reads a string and performs the required operations using separate functions.

- Convert a lowercase string to uppercase without using built-in string conversion functions.
- Reverse a string.
- Check whether a string is a palindrome.
- Continue displaying the menu until the user selects Exit.

2. Standard Test Cases

The test cases include all required inputs: menu choice and the complete input string.

TC	Choice	Input String	Expected Result
TC1	1	hello	HELLO
TC2	1	programming	PROGRAMMING
TC3	2	computer	retupmoc
TC4	2	hello	olleh
TC5	3	madam	Palindrome
TC6	3	hello	Not a palindrome
TC7	4	hello	Program exits

3. Algorithm

A. Common Menu Logic

1. Start.
2. Read a string from the user.
3. Display the menu options.
4. Read the user choice.
5. Use switch-case and call the corresponding function.
6. Repeat the menu until the user selects Exit.
7. Stop.

B. Convert Lowercase to Uppercase

1. Start from the first character of the string.
2. Check each character until the null character is reached.
3. If a character is between a and z, subtract 32 from its ASCII value.
4. Store the converted character back in the string.
5. Display the uppercase string.

C. Reverse a String

1. Find the length of the string using a loop.

2. Set i to 0 and j to length - 1.
3. Swap the characters at positions i and j.
4. Increase i and decrease j until i is not less than j.
5. Display the reversed string.

D. Check Palindrome

1. Find the length of the string using a loop.
2. Compare the first character with the last character.
3. Continue comparing characters while moving toward the center.
4. If any pair is different, the string is not a palindrome.
5. Otherwise, the string is a palindrome.

4. C Program

```
#include <stdio.h>
```

```
void toUppercase(char str[])
```

```
{
    int i = 0;
    while (str[i] != '\0')
    {
        if (str[i] >= 'a' && str[i] <= 'z')
        {
            str[i] = str[i] - 32;
        }
        i++;
    }
    printf("Uppercase string = %s\n", str);
}
```

```
void reverseString(char str[])
```

```
{
    int i = 0, j = 0;
    char temp;
    while (str[j] != '\0')
    {
        j++;
    }
    j--;
    while (i < j)
    {
```

```

    temp = str[i];
    str[i] = str[j];
    str[j] = temp;
    i++;
    j--;
}
printf("Reversed string = %s\n", str);
}
void checkPalindrome(char str[])
{
    int i = 0, j = 0, flag = 1;
    while (str[j] != '\0')
    {
        j++;
    }
    j--;
    while (i < j)
    {
        if (str[i] != str[j])
        {
            flag = 0;
            break;
        }
        i++;
        j--;
    }
    if (flag == 1)
    {
        printf("The string is a palindrome.\n");
    } else
    {
        printf("The string is not a palindrome.\n");
    }
}
int main()

```

```
{
    char str[100];
    int choice;
    printf("Enter a string: ");
    scanf(" %99[^\n]", str);
    do
    {
        printf("\nMENU\n");
        printf("1. Convert lowercase string to uppercase\n");
        printf("2. Reverse the string\n");
        printf("3. Check palindrome\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                toUppercase(str);
                break;
            case 2:
                reverseString(str);
                break;
            case 3:
                checkPalindrome(str);
                break;
            case 4:
                printf("Exiting program.\n");
                break;
            default:
                printf("Invalid choice.\n");
        }
    }
    while (choice != 4);
    return 0;
}
```

5. Compilation and Execution Evidence

```
kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)~/Desktop
$ gedit question.c

** (gedit:2075): WARNING **: 13:01:10.945: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:2075): WARNING **: 13:01:10.945: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

(kali@kali)~/Desktop
$ gcc question.c -o question

(kali@kali)~/Desktop
$ ./question
Enter a string: hello

MENU
1. Convert lowercase string to uppercase
2. Reverse the string
3. Check palindrome
4. Exit
Enter your choice: 1
Uppercase string = HELLO

MENU
1. Convert lowercase string to uppercase
2. Reverse the string
3. Check palindrome
4. Exit
Enter your choice: 2
Reversed string = OLLEH

MENU
1. Convert lowercase string to uppercase
2. Reverse the string
3. Check palindrome
4. Exit
Enter your choice: 3
The string is not a palindrome.

MENU
1. Convert lowercase string to uppercase
2. Reverse the string
```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
3. Check palindrome
4. Exit
Enter your choice: 4
Exiting program.

(kali@kali)~/Desktop
$ ./question
Enter a string: programming

MENU
1. Convert lowercase string to uppercase
2. Reverse the string
3. Check palindrome
4. Exit
Enter your choice: 1
Uppercase string = PROGRAMMING

MENU
1. Convert lowercase string to uppercase
2. Reverse the string
3. Check palindrome
4. Exit
Enter your choice: 4
Exiting program.

(kali@kali)~/Desktop
$ ./question
Enter a string: computer

MENU
1. Convert lowercase string to uppercase
2. Reverse the string
3. Check palindrome
4. Exit
Enter your choice: 2
Reversed string = retupmoc

MENU
1. Convert lowercase string to uppercase
2. Reverse the string
3. Check palindrome
```

```
kali@kali: ~/Desktop
Session Actions Edit View Help

(kali@kali)~/Desktop
$ ./question
Enter a string: computer

MENU
1. Convert lowercase string to uppercase
2. Reverse the string
3. Check palindrome
4. Exit
Enter your choice: 2
Reversed string = retupmoc

MENU
1. Convert lowercase string to uppercase
2. Reverse the string
3. Check palindrome
4. Exit
Enter your choice: 4
Exiting program.

(kali@kali)~/Desktop
$ ./question
Enter a string: madam

MENU
1. Convert lowercase string to uppercase
2. Reverse the string
3. Check palindrome
4. Exit
Enter your choice: 3
The string is a palindrome.

MENU
1. Convert lowercase string to uppercase
2. Reverse the string
3. Check palindrome
4. Exit
Enter your choice: 4
Exiting program.
```

6. Verification of All Test Cases

TC	Choice	Input String	Actual Result	Status
TC1	1	hello	HELLO	Pass
TC2	1	programming	PROGRAMMING	Pass
TC3	2	computer	retupmoc	Pass
TC4	2	hello	olleh	Pass
TC5	3	madam	Palindrome	Pass
TC6	3	hello	Not a palindrome	Pass
TC7	4	hello	Program exits	Pass